

Prometheus Field Journal – R001 & R002

A narrative research log preserving the emergence of the experiments rather than only the conclusions.

Origin

The work began with a simple question: can pine needles be converted into char instead of ash using a sealed paint can heated in a propane BBQ?

R001 – Pine Needles

Feedstock consisted primarily of pine needles in a vented paint can. Observations included smoke onset, increasing smoke, decline of smoke, and eventual cessation. The resulting material was black, brittle, structurally recognizable, and retained the geometry of the original needles. Water tests followed. The can walls accumulated only modest residue. A surprise pine cone fragment accompanied the run and became an accidental comparison specimen.

Needle Char Observations

Needles retained recognizable structure, fractured rather than bent, floated initially in water, and produced only lightly colored water. Char appeared porous and carbon-rich rather than ash-like.

Emergent Questions

Would denser packing increase yield? Would cones behave differently? Could smoke become a process signal? Could the smoke itself be burned as fuel?

R002 – Pine Cones

The can was scrubbed and reloaded with dense-packed pine cones. Gross loaded weight was approximately 2.29 lb. Ignition occurred at 16:33. First smoke appeared at 16:50, indicating a significantly slower thermal ramp than the needle run.

Temporal Topology

R001 required roughly 25 minutes. R002 required roughly 47–57 minutes before shutdown. Cones appeared to produce a longer and smoother volatile-release curve than needles. This suggested that feedstock geometry and density affect process timing.

Smoke as Signal

A central insight emerged: smoke is not merely a byproduct. Smoke became evidence of volatile release, then a candidate fuel source, and finally a candidate sensor. Future designs may use an afterburner flame or temperature measurement instead of visible smoke as the completion signal.

Tetrahedral Retort

Discussion evolved toward a tetrahedral retort geometry. The hypothesis: increased exposed surface area, controlled vapor collection at an apex, improved heat transfer, and a shorter process topology compared with a cylindrical vessel.

Propane and Fire Safety

A notable observation was the absence of flying embers. The propane-heated retort appeared fundamentally different from open burning. This may become important in future discussions of safe testing during fire season.

Yield and Morphology

Cone char remained black throughout, retained cone structure, and appeared substantially denser and higher-yield than needle char. A small glowing ember appeared when oxygen was introduced too early, reinforcing the importance of sealed cooling.

Hydrology

Needle char was soaked overnight. Water coloration remained surprisingly light. Follow-on measurements were planned to examine water retention, brittleness, and drying behavior.

Smokey Bear

A soot pattern on the grill lid unexpectedly resembled Smokey Bear. The appearance became an unofficial mascot and reminder that evidence occasionally arrives with a sense of humor.

Next Experiments

Dense mixed litter runs; cone/needle ratio optimization; afterburner concepts; thermal-mass comparisons (stainless, steel, brick); char hydrology; and automated shutdown based on temporal topology or afterburner temperature.

Closing Reflection

The day began with curiosity about pine needles. It ended with successful char production, temporal measurements, design hypotheses, safety observations, hydrology experiments, and the beginnings of a scalable Prometheus architecture. Reality repeatedly supplied the next question.